

Awni Altabaa

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Summary. My research studies the foundations of machine learning through complementary empirical and theoretical lenses, with a focus on developing novel methods and architectures in deep learning to improve systematic compositional generalization and data efficiency, sometimes drawing inspiration from biological intelligence to achieve human-like reasoning and out-of-distribution generalization. Published author in venues including NeurIPS, ICLR, ICML, and TMLR. Skilled programmer, with extensive experience in deep learning frameworks, and ability to rapidly prototype new ideas and conduct large-scale experiments.

Research Interests

Novel Neural Network Architectures · Theoretical Foundations of Modern ML Methods · Out-of-Distribution Generalization · Inductive Biases · Compositional & Relational Reasoning in Neural Networks · Reinforcement Learning · Information-Theoretic Perspectives on Learning and Reasoning · Large Language Models · LLM Agents · Connections Between Cognitive Neuroscience and Machine Intelligence

Education

- 2022 – 2027 (expected) ♦ **Ph.D. in Statistics & Data Science.** Yale University — New Haven, Connecticut.
- 2022 – 2024 ♦ **Master's in Statistics.** Yale University — New Haven, Connecticut.
- 2018 – 2022 ♦ **Bachelor's in Applied Mathematics.** Queen's University — Kingston, Ontario.

Relevant Graduate Coursework: Advanced Optimization Techniques, Measure-Theoretic Probability, Decision Theory, Theory of RL, High-dimensional Statistics, Statistical Inference on High-Dim Graph-Structured Data, Harmonic Analysis on Graphs, Statistical Learning Theory, Sampling & Optimal Transport

Overview of Recent Research

Developing architectures with enhanced relational reasoning and systematic compositional generalization.

- ♦ In our *ICLR* paper [5], we propose a novel neural module called the *Abstractor* for learning relational representations within the Transformer framework, motivated by an architectural inductive bias for relational learning that disentangles relational information from object-level features.
- ♦ Based on the ideas developed in the *Abstractor* [5], our *TICS* paper [7] with collaborators in cognitive neuroscience proposes the concept of a *relational information bottleneck* as a possible mechanisms for abstraction in the human mind and brain.
- ♦ In our *TMLR* paper [6], we propose a neural architecture based on a convolution-like operation that captures higher-order hierarchical relational patterns among groupings of objects.
- ♦ In our follow-up *ICML* work [3], we introduce the *Dual-Attention Transformer (DAT)*, a novel architecture based on Transformers that integrates sensory information (object properties) and relational information (relationships between objects) with distinct attention mechanisms. See the project webpage for more details: 🌐 www.awni.xyz/dual-attention. Open Source Implementation: 🌐 [www.pypi.org/project/dual-attention](https://pypi.org/project/dual-attention).

Theoretical Characterization of Statistical Aspects of Modern Practice in Artificial Intelligence.

- ♦ In our *NeurIPS* spotlight paper [2], we develop a statistical theory of learning under chain-of-thought (CoT) supervision, introducing an information-theoretic measure to quantify the additional discriminative power gained from observing the reasoning process, and demonstrate that CoT supervision can yield significantly faster learning rates compared to standard “end-to-end” supervision. See the project webpage for more details: 🌐 www.awni.xyz/cot-info.
- ♦ In our *NeurIPS* paper [4], we study how the structure of causal dependence between system variables affects the statistical complexity of reinforcement learning in general sequential decision-making problems, providing a more powerful model of partial observability phenomena.

Research Publications

- 1 **A. Altabaa**, S. Chen, J. Lafferty, and Z. Yang, “Unlocking out-of-distribution generalization in transformers via recursive latent space reasoning,” 2025. arXiv: 2510.14095 [cs.LG],
🌐 Project webpage: www.awni.xyz/algorithmic-generalization-transformer-architectures.
- 2 **A. Altabaa**, O. Montasser, and J. Lafferty, “CoT Information: Improved Sample Complexity under Chain-of-Thought Supervision,” *Neural Information Processing Systems (NeurIPS)*, *spotlight*, 2025,
🌐 Project webpage: www.awni.xyz/cot-info.
- 3 **A. Altabaa** and J. Lafferty, “Disentangling and integrating relational and sensory information in transformer architectures,” *International Conference on Machine Learning (ICML)*, 2025,
🌐 Project webpage: www.awni.xyz/dual-attention.
- 4 **A. Altabaa** and Z. Yang, “On the role of information structure in reinforcement learning for partially-observable sequential teams and games,” *Neural Information Processing Systems (NeurIPS)*, 2024.
- 5 **A. Altabaa**, T. W. Webb, J. D. Cohen, and J. Lafferty, “Abstractors and relational cross-attention: An inductive bias for explicit relational reasoning in transformers,” in *International Conference on Learning Representations (ICLR)*, 2024, 🌐 Project webpage: www.awni.xyz/abstractor.
- 6 **A. Altabaa** and J. Lafferty, “Learning hierarchical relational representations through relational convolutions,” *Transactions on Machine Learning Research (TMLR)*, 2024,
🌐 Project webpage: www.awni.xyz/relational-convolutions.
- 7 T. W. Webb, S. M. Frankland, **A. Altabaa**, S. Segert, K. Krishnamurthy, D. Campbell, J. Russin, T. Giallanza, R. O'Reilly, J. Lafferty, *et al.*, “The relational bottleneck as an inductive bias for efficient abstraction,” *Trends in Cognitive Sciences (TICS)*, 2024.
- 8 **A. Altabaa** and J. Lafferty, “Approximation of relation functions and attention mechanisms,” *Information Theory, Probability and Statistical Learning: A Festschrift in Honor of Andrew Barron*, 2024.
- 9 **A. Altabaa**, B. Yongacoglu, and S. Yüksel, “Decentralized multi-agent reinforcement learning for continuous-space stochastic games,” in *2023 American Control Conference (ACC)*, 2023.
- 10 **A. Altabaa**, D. Huang, C. Byles-Ho, H. Khatib, F. Sosa, and T. Hu, “Genedragnn: Gene disease prioritization using graph neural networks,” in *IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB)*, 2022.

Industry Experience

Summer 2025 ♦ **Quantitative Researcher Intern.** Citadel Securities.
Alpha research @ Systematic Equities / FICC. Project: LLM agent + Alpha Modeling.

Technical Skills

Python · Deep Learning Frameworks (PyTorch, Tensorflow, & JAX) · LLM Agent Frameworks · Version Control (Git) · High-Performance Computing (SLURM, Multi-GPU/TPU) · Efficient Distributed Training: DDP/FSDP/DeepSpeed, Mixed Precision, Profiling, Checkpointing · Language Model Pretraining, Fine-Tuning, and Evaluation · Reproducible ML: Config Mgmt, Seeding/Determinism, Unit Tests · Experiment Tracking (W&B, MLflow) · Data Visualization and Analysis (Matplotlib, Seaborn, Plotly)

Misc

Awards & Honors ♦ NSF AI Institute for Artificial and Natural Intelligence Fellow/Trainee
Review Service ♦ Journal of Data Science, NeurIPS 2024, ICLR 2025, NeurIPS 2025